

Expno: 01

AZURE DEVOPS ENVIRONMENT SETUP

DATE:

AIM:

To set up and access the Azure DevOps environment by creating an organization through the Azure Portal

INSTALLATION:

1. Open your web browser and go to the Azure website: <https://azure.microsoft.com/in-us/get-started/azure-portal>.
Sign in using your microsoft account credentials.
If you don't have a microsoft account, you can create ~~and~~ here: <https://signup.live.com/?lic=1>.
2. Azure home page
3. Open DevOps environment in the Azure platform by typing Azure DevOps organizations in the search bar.
4. Click on the my Azure DevOps Organisation link and create an ~~organisation~~ and you should be taken to the Azure DevOps Organization home page.

RESULT:

~~Successfully~~ accessed the Azure environment and ~~created~~ a new organisation through the Azure Portal.

AIM:

To Setup on Azure devops project for efficient collaboration and agile work management

Procedure:

1. Create an Azure Account

Azure devOps

sunidhi.ks.2024.it@rajalakshmi.edu.in

Name your Azure devops organizations
dev.azure.com/

2. Create the First project in your Organization

(a) After the organization is setup, you'll need to create your first project. This is where you'll begin to manage code, pipelines, work items and more

(b) On the organizations home page, click on the new project button

(c) Enter the project name, description and visibility options.

Name: choose a name for the Project

Description: optionally, add a description
provide more context about the project

Visibility:

Choose whether you want the project to be private (accessible only to those included) or public

(d) One, you've filled out the details, disk, create to setup your project

CREATE NEW PROJECT:

Project-name: Online quiz system

Visibility: Public

Version control: Git

Work item process:

Cancel Create

3. Once logged in Azure you are in the correct organization. If you're part of multiple organizations you can switch between them from the top left corner (next to your user profile). Click on the organization name and you should be taken to the Azure devOps organization home page

Srinidhi K.S

srinidhi.k.s.2024.it@rajalakshmi.edu.in
dev.azure.com/

project-online quiz system

A. Project Dashboard

Online Quiz System

About this project

The online Quiz system is a web-based application developed to conduct quizzes in an array and efficient way it automates questions management Quiz participation and result evaluation, reducing manual work. User can attend quizzes online on the end receive instant results. The system improves accuracy. Save time and supports interactive Learning through digital assessment. Overall, it provides a simple and effective platform for on the examination.

5) TO MANAGE USER STORY

(a) From the left-hand navigation menu. Click on Boards. This will take you to the main board page. Where you can manage work items, backlogs and sprints.

(b) On the work item page you'll see the option to add a work item at the top. Alternatively, you can find a button or add new work item, depending on the view you're in. From the add a work item dropdown. Select user story. This will open a form to enter details for the user story.

Boards



Work items



New-work item



User-story.



RESULT:

Successfully created an Azure DevOps project with ~~User~~ story management and Agile workflow setup.

Exp: 03

date:

SETTING-UP EPIC, FEATURES AND

USER STORY FOR PROJECT PLANNING

AIM:

To learn about how to create epic, user story, backlogs for your assigned project

PROCEDURE:

Create epic. Features. User stories, task.

1. Fill in epic.

Epic: Quiz management (Admin)

This epic includes all administration features for creating configuring and maintaining quizzes. It enables admins to manage quiz. It enables admins to manage quiz content, availability and structure

2. Fill in Features

- Hosted the online quiz system using Azure app service
- Stored quiz data and user information in Azure SQL database
- Implemented secure login using Azure, directory
- Used Azure functions to automatically calculate quiz score
- Monitored application performance using Azure monitor

3. Fill in user story details

4. As a student, I want to review my past quiz attempt so that I can track
5. As a student I want to see, my result after submission so that I can evaluate
6. As a student, I want to submit my quiz so that my answer can be evaluated
7. As a student, I want to select or enter answers so that I can respond to quiz questions
8. As a student, I want to attempt a quiz within a given time limit so that I can complete my assessment
9. As a student, I want to view available quizzes so that I can select one to attempt
10. As a student, I want to login securely so that I can access my quizzes
11. As a student, I want to create an account so that I can take quizzes

RESULT:

Thus, the creation of copies features, user story and ~~talk~~ has been created successfully.

Expno: 04

SPRINT PLANNING

DATE:

AIM:

TO assign user story to specific sprint for the online quiz system

SPRINT PLANNING:

Sprint 01:

As an admin, I want to log-in securely so that I can manage the system

Sprint 02:

→ As an admin, I want to create a new quiz so that students can attempt it.

→ As an admin, I want to add, edit or delete question so that quiz remain accurate and update.

→ As an admin, I want to view student quiz results so that I can evaluate their performance.

Sprint 03:

As a user, I want the quiz pages to load quickly so that I can start my quiz without waiting.

As a user, I want my account information and quiz data to be secure so that unauthorized user cannot access them.

As a student, I want my quiz answers to be automatically saved so that I do not lose my progress if there is a connection issue

As a student, I want a simple and clean interface so that I can easily navigate and complex my quiz.

SPRINT 04:

As an administrator, I want the system to support many students at the same time so that large quiz can be conducted without system slowdown.

As a admin, I want the quiz system work on different device and browsers so that I can access it from anywhere.

As a system admin, I want the system to be easy to maintain so that updates and improvements can be implements quickly.

RESULT:

The sprints are created for the online quiz management system.

Exp no: 05

POKER ESTIMATION

Date:

AIM:

To create poker estimation for the user stories
online quiz system

POKER ESTIMATION:

story points measure:

- Effort
- Complexity
- time

This uses fibonacci scale: 1, 2, 3, 5, 8, 13 . . .

STORY POINT FOR ONLINE QUIZ SYSTEM:

USER STORY	STORY POINTS	REASON
User registration and login	3	Authentication + input + validation
Create Quiz question	5	database operation + Form handling
Attempt quiz	5	Timer + answer selection login
view quiz result	3	store calculations + data display
Admin manage question	8	CRUD operations + Control Panel

Display user friendly interface	3	UI design and navigation
---------------------------------	---	--------------------------

Total = 81 story points

Boards → work items → user story → story points:

(1|2|3|4|5) → save

Repeat for all user stories

RESULT:

The estimation story points is created for the project online quiz system using poker estimation

EXPNO:

DESIGNING CLASS AND SEQUENCE

DATE:

DIAGRAMS FOR PROJECT ARCHITECTURE

AIM:

To design a class diagram and sequence diagram for the given Online Quiz system

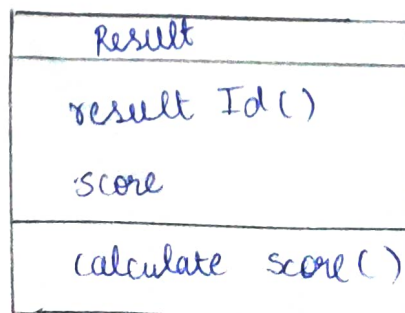
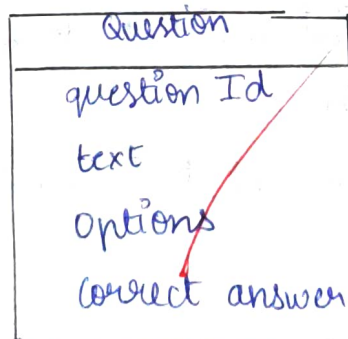
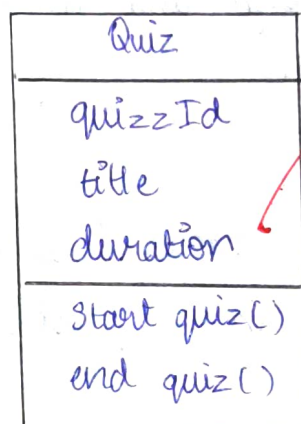
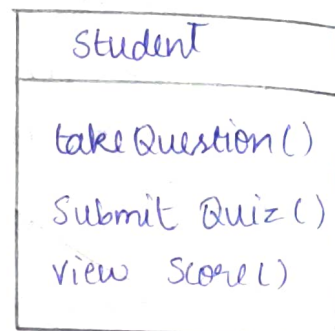
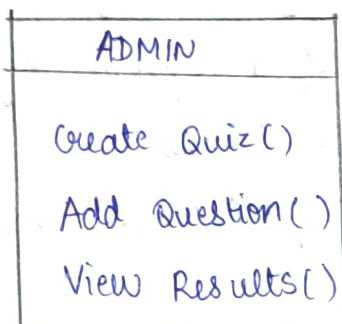
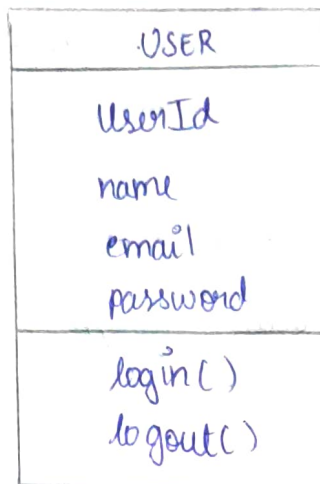
CLASS DIAGRAM:

- * A class diagram is a type of UML diagram that represents the structure of a system by showing classes. Attributes (variables). Methods (Functions), Relationships (association, inheritance, etc)
- * It is used to design and understand the system Architecture.

SEQUENCE DIAGRAM:

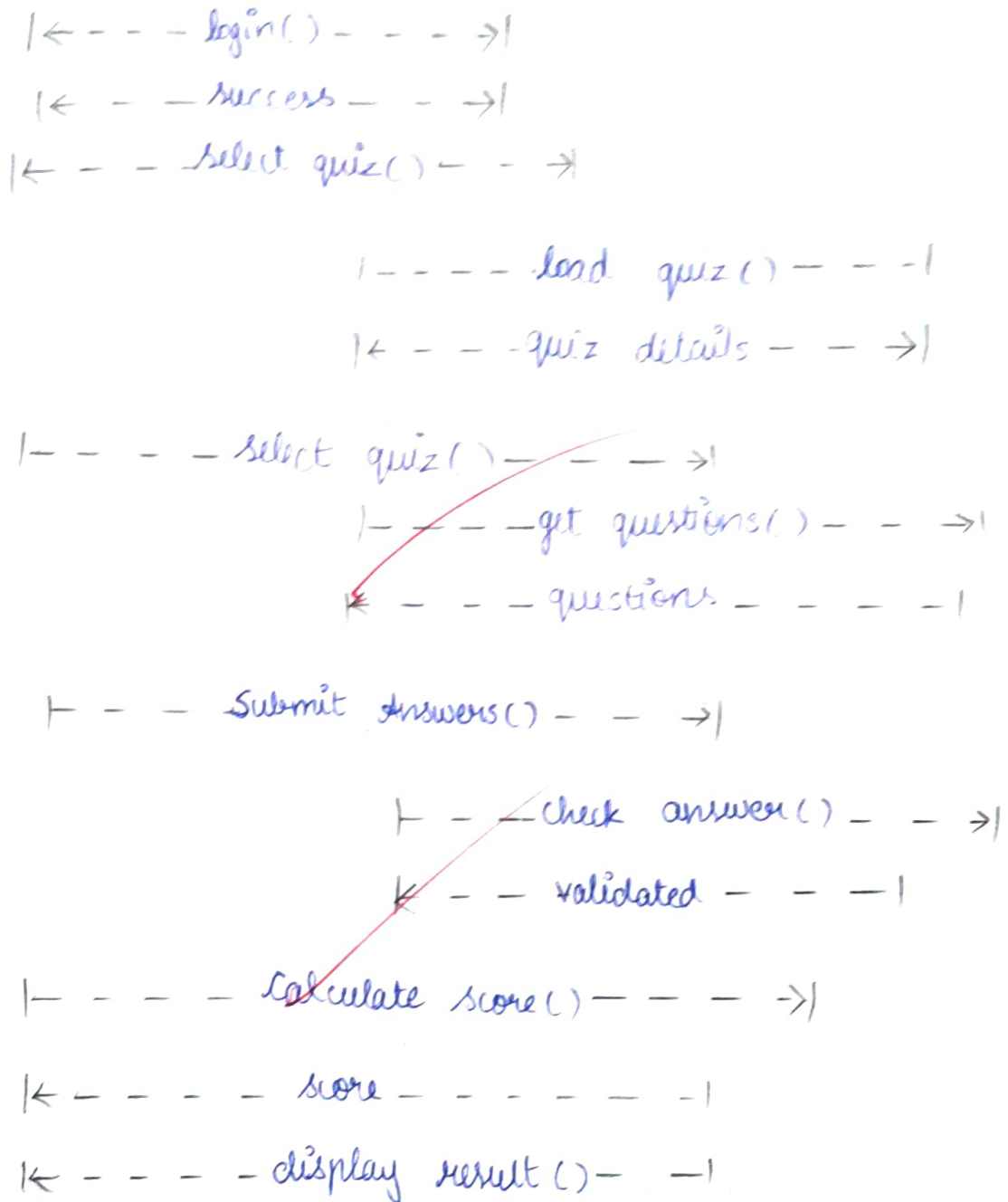
- * A sequence diagram shows how objects interact with each other in a particular sequence of time
- * It represent Object involved, Message exchange order of execution.
- * It is used to model the flow of operations step by step

Sample class diagram



Sample Sequence diagram

Student	System	Quiz
Question	Result	



RESULTS:

The class diagram and sequence diagram is ~~designed~~ successfully for the online quiz system

EXP NO: 01

DESIGNING ARCHITECTURE AND ER

DATE:

DIAGRAM FOR PROJECT STRUCTURE

AIM:

To design an Architectural diagram and ER diagram for the given project. (Online Quiz system)

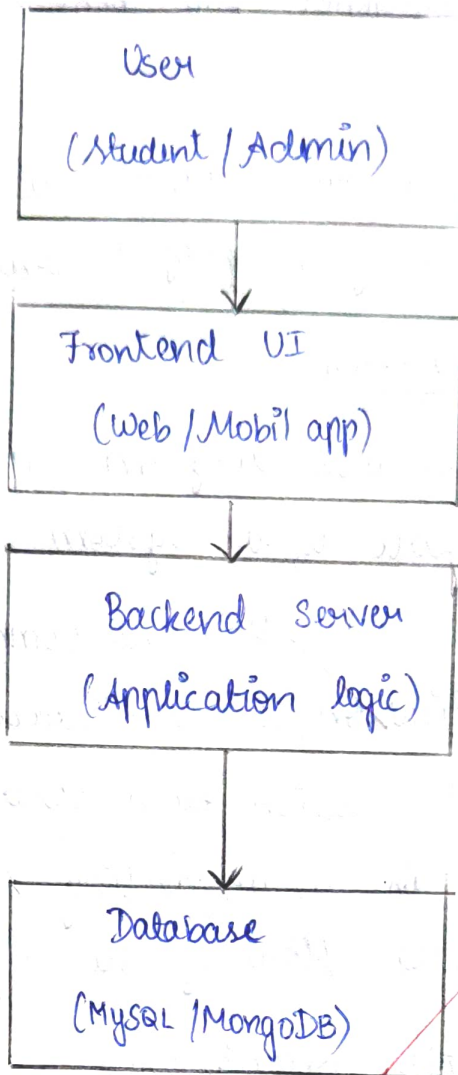
ARCHITECTURAL DIAGRAM:

- * An architectural diagram represent the overall structure of a system
- * It shows how different components like user interface Application / server, Database, External Service interact with each other.
- * It helps in understanding the high-level design and data flow of the system.

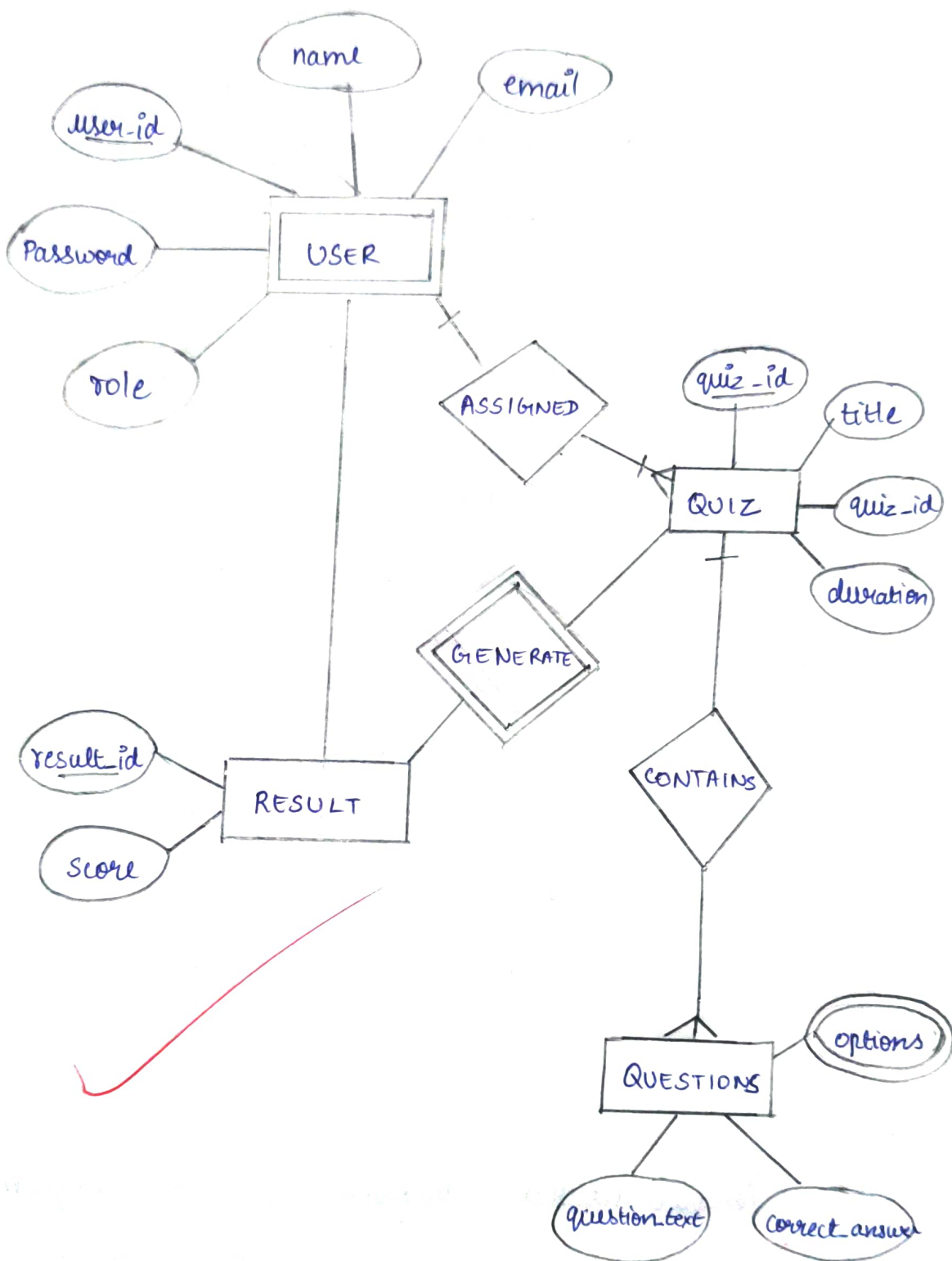
ER DIAGRAM: [Entity Relationship Diagram]

- * An ER diagram is used to represent the database structure of a system
- * It shows entities (tables), Attributes (fields), Relationships between entities
- * It helps in designing and Organizing the database schema.

ARCHITECTURAL DIAGRAM:



ER DIAGRAM: ONLINE QUIZ SYSTEM



RESULT:

The Architectural diagram and ER diagram is ~~designed~~ successfully for the online quiz system.

EXPNO: ?

TESTING - TEST PLANS AND TEST CASES

DATE:

AIM:

Test plans and test case and write two test cases for at least five user stories showcasing the happy path and various scenarios in Azure DevOps platform.

TEST PLANNING AND TEST CASE DESIGN PROCEDURE:

1. Understand core features of the Application:

- * User registration and student/teacher login
- * Quiz Creation (Setting Questions, timers and marks)
- * Taking a quiz (Real time countdown and navigation)
- * Automated grading and result generation
- * Leaderboard and Performance Analytics

2. Define user Interactions:

- * Simulate user behaviours such as answering a question
- * Submitting before the timer ends or entering incorrect login credentials

3. Design Happy path and test cases:

- * Focused on validating features under normal conditions.
- * Eg: User completes a quiz successfully and receives an instant score.

4. Design Error path test cases

- * Simulate negative scenarios to test system bushtness
- * Eg: Login fails with an unregistered email
Quiz auto submits when the internet is disconnected

5. Breakdown steps and expected results

- * Each cost includes clear actions and the specific outcome expected to ensure clarity for manual and automated testing

6. User clear naming & IOS

- * Test cases are identified as root success registration

- * Tcos - Invalid code, etc.

7. Separate Test suits

- * Grouped by functionality: Authentication, Quiz result processing

8. Prioritize and Review

- * Critical actions like quiz submission are marked as high priority

Test cases for Online quiz system.

Test ID	User Story
TC 01	User login
TC 02	User login
TC 03	Creating quiz
TC 04	Attempting quiz
TC 05	Attempting quiz
TC 06	Result entry.

TEST SUITES:

Test suit: T801 - User Authentication (ID:101)

1. TC01 - successful sign up

- * Go to the sign-up page.
- * Enter valid name - student email and password.
- * click "sign up"

Expected Results:

- * Fields accept values without errors.
- * Account is created and user is redirected to student dashboard
- * Type - Happy path.

2. TC02 - Login with wrong password

Action:

- * Go to login page
- * Enter a registered email and an incorrect password.

* Click "Login"

Expected Results:

* Input is accepted

* Error message "Invalid username or password" is displayed

* Type - "Error path"

Test Suite : TS02 - View Available Quizzes

1. TC03 - view quiz dashboards

Action:

* login successfully.

* Navigate to my quizzes section.

Expected Results:

* All assigned and active quizzes are displayed clearly with due dates.

* type : Happy path.

2. TC04 - Quiz list loading failure

Action:

* disconnect from the internet

* Navigate to "My Quizzes"

Expected Results

* Network is efficient

* Error message "unable to load quiz list,"

"please check your connection" is shown.

* Type : Error path.

Test Suite T503 - Real time quiz timer

1. TC05 - timer display and sync

Action:

- * Start a timed quiz.
- * Observe the timer countdown in the header

Expected Result:

- * Remaining time is displayed and updates every second in real-time.
- * type: Happy-Path

2. TC06 - timer expiry Action.

Action:

- * Stay on the quiz page until the timer reaches zero. 00:00

Expected Results

- * The quiz is automatically locked / submitted
- * Message "Time has expired. your answers have been saved" is shown

Type: Happy path / System Rule.

Test Suite T504: Quiz creation and editing

1. TC07 - Create quiz with blank title

Action:

- * Navigate to "Create Quiz"
- * leave the title field blank
- * Click "Save"

Expected Results:

- * Error message "quiz title cannot be empty" is shown

- * Type: Error path

2. TC08 - Edit quiz question successfully

Action:

- * select an existing quiz

- * Click "Edit Question"

- * Modify the text and click "update"

Expected Result:

- * Question content updates successfully across the database

- * Type: Happy path

User stories:

- * As a student, "I want to sign up and login securely, so that I can access my assigned quizzes".

- * As a student, I need to see my available quizzes in the one place

- * As an instructor, I should be able to create a new quiz as needed.

* As an instructor, I should be able to edit, delete and change quiz setting

* As a student, I need to see real time remaining during a quiz.

RESULT:

The test plan and test cases for the user stories are created in Azure DevOps with happy path and error path.

EXP: 09

LOAD TESTING AND CI/CD PIPELINE

DATE:

AIM:

To perform load testing on the Quiz generator web application to evaluate its performance under multiple user, and to implement a CI/CD pipeline for automating the build and testing process of the application.

LOAD TESTING:

- * The load testing was performed on the Quiz generator web application development using flask & MongoDB.
- * Multiple virtual users were simulated to access the application simultaneously, including login, quiz creation, and quiz attempts.

PIPELINE SETUP

- * A CI/CD pipeline was configured using Azure DevOps by connecting the GitHub repository containing the Quiz generator quiz project.
- * The pipeline automates the process of installing dependencies & running the application.

YAML

trigger:

- main

Pool:

vm Image: 'ubuntu - latest'

steps:

- checkout: self

- task: UsePythonVersion@0

inputs:

versionSpec: '3.x'

displayName: 'Set up python'

- script: |

python -m pip install --upgrade pip

pip install -r requirements.txt

displayName: 'Install dependencies'

- script: |

python app.py

displayName: 'Run Flask Application'

RESULT:

The Quiz generation successfully handled multiple user requests during load testing with stable performance. The CI/CD pipeline was successfully implemented, automating the installation of dependencies and execution of the application.

EXP:10

date:

GITHUB PROJECT STRUCTURE AND NAMING CONVENTIONS FOR

AIM:

To organize the Quiz generator project using a proper folder structure and naming conventions in GitHub, making the project easy to understand, maintain & extend.

GitHub Usage

The project was uploaded to github repository for version control and collaboration. The repository includes all source code, structure and documentation, allowing easy access and modification.

PROCEDURE:

1. Created the Quiz generator project using Flask & MongoDB.
2. Organized files into folders such as templates, static and backend files.
3. Followed proper naming conventions for files & folders.
4. Created a Github repository.
5. Uploaded the project files to Github.
6. Verified the structure & readability of the project in the repository.

RESULT:

The Quiz generator project was successfully organized using a clear folder structure and naming conventions. The Github repository clearly displays the project structure, making it easy for users and developers to understand and navigate the codebase.